

Student Assessment Information Sheet

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Assessment Information	
Module Name:	GPU Programming and real-time rendering
Module Code:	CS51012
Assessment Title:	Assignment: Individual Program
Assessment Weighting:	60%
Assessment Due Date	PART1: Light touch review: week 8 (Nov 14 th), 10pm FINAL SUBMISSION: Week 11 (Dec 5 th , 10pm)
Assignment type	Individual

If you have questions about this assessment, please contact i.martin@dundee.ac.uk

Learning Outcomes
This assessment addresses the following learning outcomes of the module: - Demonstrate the use of lighting, shading and other techniques to create realistic scenes. - To develop interactive graphical software and animated sequences. - To develop and apply problem-solving, time-management, self-assessment and independent study skills.

AI Usage Guidance		
Given the broad and evolving nature of these tools, it is important to be clear about what is permitted in each assessment. Your module lecturers will indicate to what extent generative AI can be used. If you are unsure whether the use of a specific AI tool is permitted, please contact your module coordinator for guidance. Misuse or failure to disclose AI assistance may be treated as academic misconduct and subject to procedures outlined in the academic misconduct by students code of practice. The levels of AI usage in assessment are:		
RAG Rating	Permitted Level	Description
Red (Forbidden)	Yes/No	AI tools must not be used for this assessment. This includes generating content, conducting analysis, or providing answers to specific questions.
Amber (Restricted)	Yes	AI tools may be used with caution, but their use must be explicitly acknowledged. For example, AI assistance in brainstorming, drafting, or structuring ideas is allowed, but the final submission must be your own work. Any AI-generated content must be fully reviewed, edited, and critically evaluated.
Green	Yes/No	AI tools are permitted for this assessment without

(Permitted)	restriction, but their use must be explicitly acknowledged. You may use AI systems for research assistance, content generation, and other purposes, provided that the usage aligns with the learning outcomes and your understanding is demonstrated. You should engage critically with output and apply your own judgement.
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Assessment Task - ACTION SCENE with BACKGROUND in two parts

Select a real-world (3D) Action Scene with multiple moving parts and write an OpenGL program which models this Action Scene with lighting, texture, user camera controls and action (significant movement of object(s) in the scene), a background appropriate for the scene, one or more imported external objects, plus any other GPU programming techniques that could add to your action scene.

You should implement this assignment in two parts, submitting PART 1 for light-touch review/feedback.

PART 1 (20% of final grade)

For PART 1, focus on object animation with basic object shapes, lighting, camera and user controls. Please ensure that your PART 1 submission includes the following aspects:

- Two or more 3D objects with moving parts that represent a real-world object or scene.
- Lighting, appropriate for the scene.
- Movable, user-controlled camera (view position), appropriate for your scene.
- Perspective projection
- Appropriate movement of objects in your scene, with movement speed adjustable through user controls.
- You can build the scene from basic objects provided and/or include your own.
- You **MUST** not use the deprecated immediate mode, i.e. you must use vertex buffers, uniform transformation and include at least one vertex and fragment shader.
- You do not need to include a background at this stage – that is for PART 2, after we've covered different ways of doing that in class.

Please submit this as **PART 1** of this assignment. Light-touch feedback and a PART 1 grade will be given on this section of the assignment.

PART 2 (80% of final grade)

Please extend your PART 1 submission to include the following:

- Background or context appropriate to your scene. For example, an outdoor scene should show the ground and the surrounding terrain or vegetation, an indoor scene should show walls, windows etc.
- Your scene should include one or more externally created graphics objects that you have loaded into your application and rendered with appropriate lighting.
- You should include some form of texturing in your scene.
- At least one 'advanced' concept that you have learnt that can enhance your scene. Advanced means more than basic lighting and texturing in this context. Possible examples include; procedural textures, environment mapping, shadow casting, normal-mapping, particle animation, geometry or compute shaders etc. (don't feel limited by this list!).

Additional Remarks

- You are free to make use of any of the example programs on MyDundee for this class. This does not need to be referenced because I will be able to judge how much you have used and extended this. i.e., feel free to start with one of the

class examples and extend it to your purpose.

- You are also free to update your PART 1 submission while working on extending it for PART 2. i.e. please act on any feedback received to make your assignment better!
- You are also free to use pieces of code obtained from books/Internet, but these must be clearly referenced by being marked as such in source files and full references and explanations given in the accompanying report. No credit will be given for this code, but credit will be given for adapting or modifying external code as long as it is clearly referenced. Please ask me if in doubt.
- The learning outcomes being assessed for PART 1 are focused on understating how to transform and move 3D objects in a appropriate way for the scene and include lighting and user controls.
- The learning outcomes being assessed for PART 2 are focused on extending your PART1 submission through understating how to appropriate include texture, background and other more advanced graphical and GPU programming techniques.
- The mark for this assignment will be based mainly on the implementation of the Graphics functionality and realism of the program, with some credit also for its design and structure.
- Collaboration is not permitted on this assignment.
- Please remember to structure and comment your code sensibly. E.g. name and date on all source files, a comment describing each function and other comments describing any tricky section of code. Marks will be deducted if standard good programming practices aren't followed.
- Extra credit will be given for simulating a scene that moves in a realistic manner, good camera controls and lighting effects, good use of GPU shaders and implementation of appropriate graphical programming techniques.

PART 1: Deliverables

1. Your executable in a separate folder with all other files needed to run your exe. I'd like to be able to just click on it and it should run. Please test this on a different PC to your development PC before submitting it! Please include clear instructions to run your program if it requires anything further than running an executable. Please speak to me in the lab if you have any difficulties with this.
2. A short video to show your program running. This is mainly so that I can see what Graphics you created on your system, in case there are any issues that cause them to be different when I run them. This can be a link to a video and please make it no longer than 1 minute (shorter will be fine!). If you submit a link, please make it unlisted and not private, so that I can access to the link.
3. Your OpenGL program source code folder (including the full, cleaned, IDE project or make file). Please check for and delete the large temporary Intellisense files. Check the size of your zip file and if it is large, find and delete the temporary files (ask me for help with this!)

PART 2: Deliverables

1. Your executable in a separate folder with all other files needed to run your exe. I'd like to be able to just click on it and it should run. Please test this on a different PC to your development PC before submitting it! Please include clear instructions to run your program if it requires anything further than running an executable. Please speak to me in the lab if you have any difficulties with this.
2. A short video to show your program running. This is mainly so that I can see what Graphics you created on your system, in case there are any issues that cause them to be different when I run them. This can be a link to a video and please make it no longer than 1 minute (shorter will be fine!). If you submit a link, please make it

unlisted and not private, so that I can access to the link.

3. Your OpenGL program source code folder (including the full, cleaned, IDE project or make file). If using Visual Studio, please check for and delete the large temporary Intellisense files (*.ipch). Check the size of your zip file and if it is large, find and delete the temporary files (ask me for help with this!).
4. A short document containing a brief description of your design, noting any areas that you had, problems with and anything extra that you accomplished that hasn't been covered in the lectures or has been implemented particularly well. Include references to any tutorials or articles you read for help or used. Also include a section on your report to describe and justify the implementation of the advanced technique (or techniques) that you have used and suggest alternative techniques (e.g. research them yourself) that could be used to improve your scene, and suggest how they could be implemented in your application.

How to Submit

Assessments should be submitted to My Dundee in the 'Assessment and Feedback' area. Please collate all the files in your submission into a single ZIP file and upload the ZIP file to My Dundee.

Please name your ZIP file accordingly in a zip file with your name. E.g., lastname_firstname_part1.zip (for PART 1) and lastname_firstname_part2.zip

Assessment Feedback

Feedback on PART 1 of this assessment will be provided via **My Dundee** and will be available within **two weeks** of the submission deadline. Any alteration to this will be communicated with you in advance through My Dundee. A grade will be given for PART 1, representing 20% of the final grade for the whole assignment.

Feedback on PART 2 of this assessment will be provided via **My Dundee** and will be available within **three weeks** of the submission deadline. Any alteration to this will be communicated with you in advance through My Dundee. A grade will be given for PART 2, representing 20% of the final grade for the whole assignment.

Feedback can be provided in multiple ways. This can include informally during lab and tutorial sessions (e.g. discussing work and sample solutions being explored), and formally after marking (e.g. written, oral, video-based).

Academic Integrity

By submitting this Assessment, you are bound by the University of Dundee policies in relation to Academic Standards and Integrity. Please refer to [Policy Statements, Regulations and Guidance](#) for more information

Late Penalties

One grade point per day late (meaning if a submission is one day late and marked as a C2 it will receive a C3 grade). A day is defined as each 24-hour period following the submission deadline including weekends and holidays. Assessments submitted more than 5 days after the agreed deadline will receive a zero mark (AB).

Generic Assessment Criteria as applied to the Graphics assignments		
A1 - A5	Meets learning outcomes	The submission must meet all of the assignment criteria with no major errors and should demonstrate a deeper understanding of the topic by including some innovation or some additional features or improvements that have been sourced from elsewhere which demonstrates your ability to read around the subject or extend topics using your own initiative (e.g. improved camera controls, lighting effects, complex and realistic object movement etc.). All use of additional sources (books, examples on the internet, outside tutorials etc.) are encouraged but must be properly referenced.
B1 - B3		The submission should meet all of the assignment criteria with no major errors. It may contain some minor errors or areas that could be improved but, overall, all aspects of the submission should have been completed to a high a standard. The submission should demonstrate a good understanding of the topics and learning outcomes.
C1 - C3		The submission should meet all the assignment criteria or most of the criteria. There may be some omissions or errors, but these should not be so significant that they result in an incomplete or incoherent submission. The submission should demonstrate a reasonable understanding of the topics.
D1 - D3		The assignment should meet some of the assignment criteria but will contain some major errors or omissions but should still compile and run. There will be significant problems with either implementation or understanding. However, the submission should still demonstrate some understanding of the topics and learning outcomes.
MF1 - MF3	Fails to meet learning outcomes	Does not meet the required outcomes. Shows a limited understanding of relevant theories and concepts, and implementation issues. Some relevant content may be included, but there are significant gaps in the assignment criteria, fundamental errors and misunderstandings are likely. And there may be some compilation or build issues. Limited use of learning resources, implementation and presentation fall short of accuracy and proficiency.
BF - CF		Does not meet outcomes. Demonstrates a lack of understanding of relevant theories, concepts, and implementation issues. Minimal research and use of established methodologies, leading to incomplete knowledge of the subject. Significant errors and missing assignment criteria are apparent. Little to no use of learning resources. The implementation is weak and there may be compilation and build issues.